Zithronam

Powder for Oral Suspension 200mg/5ml

Name and Strength of Active Substance(s)
Each 5 ml reconstituted suspension contains Azithromycin Dihydrate USP 209.62mg eq. to Azithromycin 200mg.

Product Description
Before reconstitution: A white to off white free flowing granular powder, free from any visible foreign particles filled in HDPE bottle and sealed with SQUARE printed white color cap.

For pack size 30ml,
After reconstitution with 20ml water, it gets a white to off-white color suspension having flavored sweet taste with slightly after bitter taste.

For pack size 50ml,
After reconstitution with 35ml water, it gets a white to off-white color suspension having flavored sweet taste with slightly after bitter taste.

Pharmacodynamics
Azithromycin is the first of a subclass of macrolides antibiotics, known as azalides, and is chemically different from erythromycin. Chemically it is derived by insert ion of a nitrogen atom into the lactone ring of erythromycin A. The mode of action of azithromycin is inhibition of protein synthesis in bacteria by binding to the 50s ribosomal subunit and preventing translocation of peptides.
Azithromycin demonstrates activity in vitro against a wide range of bacteria including:

<u>Gram-positive Aerobic Bacteria</u>
Staphylococcus aureus, Streptococcus pyogenes (group A beta-hemolytic streptococci), *Streptococcus pneumoniae, alpha-hemolytic streptococci* (viridans group) and other streptococci, and *Corynebacterium diphtheriae*. Azithromycin demonstrates cross-resistance with erythromycin-resistant Gram-positive strains, including *Streptococcus faecalis* (enterococcus) and most strains of methicillin- resistant staphylococci.

<u>Gram-negative Aerobic Bacteria</u>
Haemophilus influenzae, Haemophilus parainfluenzae, Moraxella catarrhalis, Acinetobacter species, *Yersinia* species, *Legionella pneumophila*, *Bordetella pertussis, Bordetella parapertussis, Shigella species, Pasteurella species, Vibrio cholerae* and *parahaemolyticus, Plesiomonas shigelloides*. Activities against *Escherichia coli, Salmonella enteritidis, Salmonella typhi*, *Enterobacter* species, *Aeromonas hydrophila* and *Klebsiella* species are variable and susceptibility tests should be performed. *Proteus* species*, Serratia* species, *Morganella* species, and *Pseudomonas aeruginosa* are usually resistant.

<u>Anaerobic Bacteria</u>
Bacteroides fragilis and *Bacteroides* species, *Clostridium perfringens*, *Peptococcus* species and *Peptostreptococcus* species, *Fusobacterium necrophorum* and *Propionibacterium acnes*.

<u>Organisms of Sexually Transmitted Diseases</u>
Azithromycin is active against Chlamydia *trachomatis* and also shows good activity against *Treponema pallidum*, *Neisseria gonorrhoeae*, and *Haemophilus ducreyi*.

Other Organisms

Borrelia burgdorferi (Lyme disease agent), *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, *Mycoplasma hominis, Ureaplasma urealyticum, Campylobacter* species and *Listeria monocytogenes*.

Opportunistic Pathogens Associated with HIV Infections

Mycobacterium avium-intracellulare complex, Pneumocystis carinii and *Toxoplasma gondii*.

Commonly susceptible species:

Aerobic Gram-positive bacteria:

Staphylococcus aureus, *Streptococcus agalactiae*, Streptococci (Groups C, F, G) and Viridans group streptococci.

Aerobic Gram-negative bacteria:

Bordetella pertussis, *Haemophilus ducreyi*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Legionella pneumophila*, *Moraxella catarrhalis* and *Neisseria gonorrhoeae*.

Other

Chlamydia pneumoniae, *Chlamydia trachomatis*, *Mycoplasma pneumonia* and *Ureaplasma urealyticum*.

Species for which acquired resistance:

Aerobic Gram-positive bacteria: *Streptococcus pneumoniae, Streptococcus pyogenes*

Note: Azithromycin demonstrates cross-resistance with erythromycin-resistant gram-positive strains.

Inherently resistant organisms: Enterobacteriaceae, *Pseudomonas*

Pharmacokinetics

Absorption:

Following oral administration, bioavailability is approximately 37%. The time taken to peak plasma levels is 2-3 hours.

Distribution:

Orally administered azithromycin is widely distributed throughout the body. It has been demonstrated that the concentrations of azithromycin are markedly higher levels in tissue than in plasma (up to 50 times the maximum observed concentration in plasma) indicating that the drug is heavily tissue bound. Concentrations in target tissues, such as lung, tonsil and prostate exceed the MIC90 for likely pathogens after a single dose of 500 mg.

Elimination:

Plasma terminal elimination half-life closely reflects the tissue depletion half-life of 2 to 4 days. Approximately 12% of an intravenously administered dose is excreted in the urine over 3 days as the parent drug, the majority in the first 24 hours. Biliary excretion of azithromycin is a major route of elimination for unchanged drug following oral administration. Very high concentrations of unchanged drug have been found in human bile, together with 10 metabolites, formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings, and by cleavage of the cladinose conjugate. Comparison of liquid chromatography and microbiological assays in tissues suggests that metabolites play no part in the microbiological activity of azithromycin.

Pharmacokinetics in Special Patient Groups

Elderly

In elderly patients (>65 years), slightly higher AUC values were seen after a 5-day regimen than in young patients (<40 years), but these are not considered clinically significant, and hence no dose adjustment is recommended.

Renal Impairment

The pharmacokinetics of azithromycin in individual with mild to moderate renal impairment (GFR 10 - 80 ml/min) were not affected following a single 1 gram dose of immediate release azithromycin. Statistically significant differences in AUC, Cmax and CLr were observed between the group with severe renal impairment (GFR < 10 ml/min) and the group with normal renal function.

Hepatic Impairment

In patients with mild (Class A) to moderate (Class B) hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to those with normal hepatic function. In these patients urinary clearance of azithromycin appears to increase, perhaps to compensate for reduced hepatic clearance.

Indication

Azithromycin is indicated for infections caused by susceptible organisms; in lower respiratory tract infections including bronchitis and pneumonia, in skin and soft tissue infections, in acute otitis media and in upper respiratory tract infections including sinusitis and pharyngitis/tonsillitis. (Penicillin is the usual drug of choice in the treatment of *Streptococcus pyogenes* pharyngitis, including the prophylaxis of rheumatic fever. Azithromycin is generally effective in the eradication of streptococci from the oropharynx, however, data establishing the efficacy of azithromycin and the subsequent prevention of rheumatic fever are not available at present.)

In sexually transmitted diseases in men and women, azithromycin is indicated in the treatment of uncomplicated genital infections due to *Chlamydia trachomatis*. It is also indicated in the treatment of chancroid due to *Haemophilus ducreyi*; and uncomplicated genital infection due to non-multiresistant *Neisseria gonorrhoea*; concurrent infection with *Treponema pallidum* should be excluded.

Recommended Dosage

Oral azithromycin should be administered as a single daily dose. The period of dosing with regard to infection is given below. Azithromycin powder for oral suspension can be taken with or without food.

Adults

For the treatment of sexually transmitted diseases caused by Chlamydia trachomatis and Haemophilus ducreyi or susceptible Neisseria gonorrhoea, the dose is 1000 mg as a single oral dose. For patients who are allergic to penicillin and/or cephalosporins, prescribers should consult local treatment guidelines.

For all other indications in which the oral formulation is administered, the total dosage of 1500 mg should be given as 500 mg daily for 3 days. As an alternative, the same total dose can be given over 5 days with 500 mg given on day 1, then 250 mg daily on days 2 to 5.

Children
The maximum recommended total dose for any treatment is 1500 mg for children. In general, the total dose in children is 30 mg/kg. Treatment for pediatric streptococcal pharyngitis should be dosed at a different regimen (see below).

The total dose of 30 mg/kg should be given as a single daily dose of 10 mg/kg daily for 3 days, or given over 5 days with a single daily dose of 10 mg/kg on day 1, then 5 mg/kg on days 2-5. As an alternative to the above dosing, treatment for children with acute otitis media can be given as a single dose of 30 mg/kg.

For pediatric streptococcal pharyngitis, azithromycin given as a single dose of 10 mg/kg or 20 mg/kg for 3 days has been shown to be effective; however, a daily dose of 500 mg must not be exceeded. However, penicillin is the usual drug of choice in the treatment of Streptococcus pyogenes pharyngitis, including prophylaxis of rheumatic fever.

For children weighing less than 15 kg, azithromycin suspension should be measured as closely as possible. For children weighing 15 kg or more, azithromycin suspension should be administered according to the guide provided below:

Azithromycin Suspension 30 mg/kg Total Treatment Dose		
Weight (kg)	**3-Day Regimen**	**5-Day Regimen**
< 15	10 mg/kg once daily on days 1-3.	10 mg/kg on day 1, then 5 mg/kg once daily on days 2-5.
15-25	200 mg (5 ml) once daily on days 1-3.	200 mg (5 ml) on day 1, then 100 mg (2.5 ml) once daily on days 2-5.
26-35	300 mg (7.5 ml) once daily on days 1-3.	300 mg (7.5 ml) on day 1, then 150 mg (3.75 ml) once daily on days 2-5.
36-45	400 mg (10 ml) once daily on days 1-3.	400 mg (10 ml) on day 1, then 200 mg (5 ml) once daily on days 2-5.
> 45	Dose as per adults.	Dose as per adults.

Azithromycin tablets should only be administered to children weighing more than 45 kg.

Elderly
The same dosage as in adult patients is used in the elderly.
Elderly patients may be more susceptible to the development of torsades de pointes arrhythmia than younger patients.

Patients with Renal Impairment
No dose adjustment is necessary in patients with mild to moderate renal impairment (GFR 10 - 80 ml/min). Caution should be exercised when azithromycin is administered to patients with severe renal impairment (GFR < 10 ml/min).

Patients with Hepatic Impairment
The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

Mode of Administration
Oral

Contraindications
The use of this product is contraindicated in patients with a known hypersensitivity to azithromycin, erythromycin, any macrolide or ketolide antibiotic, or to any of the excipients.

Warnings and Precautions

Hypersensitivity
As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), dermatologic reactions including Stevens-Johnson Syndrome (SJS), Toxic Epidermal Necrolysis (TEN) (rarely fatal), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment.

If an allergic reaction occurs, the drug should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], Zithronam should be discontinued immediately and appropriate treatment should be urgently initiated.

Prolongation of the QT interval
Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin.
Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital or documented QT prolongation
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
- Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval

Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/ or irritability with feeding occurs.

Since liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease.

In patients receiving ergot derivatives, ergotism has been precipitated by coadministration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be co-administered.

As with any antibiotic preparation, observation for signs of superinfection with non-susceptible organisms, including fungi is recommended.

Clostridium difficile associated diarrhea (CDAD) has been reported with use of nearly all antibacterial agents, including azithromycin, and may range in severity from mild diarrhea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C difficile*.

C. difficile produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

In patients with severe renal impairment (GFR <10 ml/min) a 33% increase in systemic exposure to azithromycin was observed.

Contains Aspartame: Unsuitable for phenylketonurics.

Caution in diabetic patients: 5 ml of reconstituted suspension contains 2.151g of sucrose.

Due to the sucrose content (2.151g/5 ml of reconstituted suspension), this medicinal product is not indicated for persons with fructose intolerance (hereditary fructose intolerance), glucose-galactose malabsorption or saccharase-isomaltase deficiency.

Effects on Ability to Drive and Use Machines
There is no evidence to suggest that azithromycin may have an effect on the patient's ability to drive or operate machinery.

Interactions with Other Medicaments
Antacids
No effect on overall bioavailability was seen although peak serum concentrations were reduced by approximately 25%. In patients receiving both oral azithromycin and antacids, the drugs should not be taken simultaneously.

Cetirizine
Coadministration of a 5-day regimen of azithromycin with cetirizine 20 mg at steady-state resulted in no interaction and no significant changes in the QT interval.

Didanosine (Dideoxyinosine)
Coadministration of 1200 mg/day azithromycin with 400 mg/day didanosine in HIV-positive patients did not appear to affect the steady-state of didanosine.

Digoxin
Some of the macrolide antibiotics have been reported to impair the microbial metabolism of digoxin in the gut in some patients. In patients receiving concomitant azithromycin, a related azalide antibiotic, and digoxin the possibility of raised digoxin levels should be borne in mind.

Zidovudine
Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding is unclear, but it may be of benefit to patients.

Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot
Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Atorvastatin
Coadministration of atorvastatin (10 mg daily) and azithromycin (500 mg daily) did not alter the plasma concentrations of atorvastatin.

Carbamazepine
No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine
In a single dose of cimetidine, given 2 hours before azithromycin, , no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants
There have been reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin
Caution should be exercised before considering concurrent administration of cyclosporin and azithromycin. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz
Coadministration of a 600 mg single dose of azithromycin and 400 mg efavirenz daily for 7 days did not result in any clinically significant interactions.

Fluconazole
Coadministration of a single dose of 1200 mg azithromycin did not alter the pharmacokinetics of a single dose of 800 mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the coadministration of fluconazole, however a clinically insignificant decrease in Cmax (18%) of azithromycin was observed.

Indinavir
Coadministration of a single dose of 1200 mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800 mg three times daily for 5 days.

Methylprednisolone
Azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam
Coadministration of azithromycin 500 mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15 mg dose of midazolam.

Nelfinavir
Coadministration of azithromycin (1200 mg) and nelfinavir at steady state (750 mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin
Coadministration of azithromycin and rifabutin did not affect the serum concentrations of either drug.

Neutropenia was observed in patients receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil
There was no evidence of an effect of azithromycin (500mg daily for 3 days) on the AUC and Cmax, of sildenafil or its major circulating metabolite.

Terfenadine
Studies have reported no evidence of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline
There is no evidence of a clinically significant pharmacokinetic interaction when azithromycin and theophylline are co-administered to healthy volunteers.

Triazolam
Coadministration of azithromycin with 0.125 mg triazolam had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim/sulfamethoxazole
Coadministration of trimethoprim/sulfamethoxazole DS (160 mg/800 mg) for 7 days with azithromycin 1200 mg on Day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other.

Cisapride
Cisapride is metabolized in the liver by the enzyme CYP 3A4. Because macrolides inhibit this enzyme, concomitant administration of cisapride may cause the increase of QT interval prolongation, ventricular arrhythmias and torsades de pointes.

Astemizole, alfentanil
Caution is advised in the co-administration of astemizole or alfentanil with Azithromycin because of the known enhancing effect of these medicines when used concurrently with the macrolide antibiotic erythromycin.

Statement on Usage during Pregnancy and Lactation
Pregnancy:
The safety of azithromycin has not been confirmed with regard to the use of the active substance during pregnancy. Azithromycin should be used during pregnancy only if clearly needed.

Breast-feeding:
A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from azithromycin therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Adverse Effects / Undesirable Effects
Azithromycin is well tolerated with a low incidence of side effects.

Blood and Lymphatic System Disorders
Transient episodes of mild neutropenia have occasionally been observed, although a causal relationship to azithromycin has not been established.

Ear and Labyrinth Disorders
Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported in some patients receiving azithromycin. Many of these have been associated with prolonged use of high doses. In those cases where follow-up information was available the majority of these events were reversible.

Gastrointestinal Disorders
Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps), and flatulence.

Hepatobiliary Disorders
Abnormal liver function.

Skin and Subcutaneous Tissue Disorders
Allergic reactions including rash and angioedema.
Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

General Disorders and Administration Site Conditions
Local pain and inflammation at the site of infusion.

In post-marketing experience, the following additional undesirable effects have been reported:
Infections and Infestations

Moniliasis and vaginitis.

Blood and Lymphatic System Disorders
Thrombocytopenia.

Immune System Disorders
Anaphylaxis (rarely fatal)

Metabolism and Nutrition Disorders
Anorexia.

Psychiatric Disorders
Aggressive reaction, nervousness, agitation, and anxiety.

Nervous System Disorders
Dizziness, convulsions (as seen with other macrolides), headache, hyperactivity, hypoesthesia, paresthesia, somnolence, and syncope. There have been rare reports of taste/smell perversion and/or loss. However, a causal relationship has not been established.

Ear and Labyrinth Disorders
Vertigo.

Cardiac Disorders
Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes (see **Special Warnings and Precautions for Use)**.

Skin and Subcutaneous Tissue Disorders
Allergic reactions including pruritus, rash, photosensitivity, edema, urticaria, and angioedema. Rarely, serious cutaneous adverse reactions including erythema multiforme, SJS, TEN and DRESS have been reported.

Vascular Disorders
Hypotension

Gastrointestinal Disorders
Infantile hypertrophic pyloric stenosis
vomiting/diarrhea (rarely resulting in dehydration) dyspepsia, constipation, pseudomembranous colitis, pancreatitis, and rare reports if tongue discoloration.

Hepatobiliary Disorders
Hepatitis and cholestatic jaundice have been reported, as well as rare cases of hepatic necrosis and hepatic failure, which have rarely resulted in death. However, a causal relationship has not been established.

Musculoskeletal and Connective Tissue Disorders
Arthralgia.

Renal and Urinary Disorders
Interstitial nephritis and acute renal failure.

General Disorders and Administration Site Conditions
Asthenia has been reported, although a causal relationship has not been established; fatigue, and malaise.

Overdose and Treatment
Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. The typical symptoms of an overdose with macrolide antibiotics include reversible loss of hearing, severe nausea, vomiting and diarrhoea. In the event of overdose, the administration of medicinal charcoal and general symptomatic treatment and supportive measures are indicated as required.

Instructions for Use and Handling, and Disposal
Zithronam powder for oral suspension can be taken with or without food.

(Pack size 30ml)
Tap the bottle to loosen the powder. Then add 20 ml of boiled and cooled water to the content of the bottle and shake well to mix uniformly. Shake immediately prior to use.

(Pack size 50ml)
Tap the bottle to loosen the powder. Then add 35 ml of boiled and cooled water to the content of the bottle and shake well to mix uniformly. Shake immediately prior to use.

For children weighing less than 15 kg, the suspension should be measured as closely as possible.
For children weighing 15 kg or more, the suspension should be administered using an appropriate measuring device.

Storage Conditions
Zithronam powder for oral suspension should be stored below 30°C. Protect from light and moisture.
After reconstitution the suspension should be stored below 30°C and be used within 5 days.

Dosage Forms and Packaging Available
Powder for Oral Suspension. Pack in HDPE round white bottle, pack size of 30ml and 50ml.

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